

REVIEW

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Advancing inclusive smart learning environments: a systematic review of technologies, challenges, and best practices for students with disabilities

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Abstract

This systematic literature review (SLR) explores the roles of Smart Learning Environments (SLEs) in fostering inclusive education for students with disabilities, with a focus on their applications, challenges, and best practices. Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology, this study conducts a rigorous analysis of peer-reviewed articles and conference proceedings from six academic databases. Our study identifies current applications and key technological advancements that enhance accessibility and learning outcomes for students with disabilities. By synthesizing current findings, it also addresses systemic barriers faced by learners with disabilities when engaging with SLEs. It also provides actionable insights for designing and implementing SLEs that promote equity, inclusivity, and empowerment in diverse educational contexts. From an initial pool of 2297 sources published between 2019 and 2024, a rigorous screening and eligibility process resulted in the selection of 43 studies. These studies provide valuable insights into how SLEs can address the diverse needs of students with disabilities while highlighting persistent barriers and areas for improvement. Our study demonstrates the transformative potential of SLE in redefining education for students with disabilities. While these advancements improve learning experiences, ongoing challenges restrict their potential. To bridge these gaps, universal design must be implemented, as well as strengthened regulatory frameworks and targeted professional development. Future research should focus on large-scale, longitudinal studies to examine the long-term impact of SLEs and make significant progress toward truly inclusive education.

Keywords: Smart learning environments, Inclusive education, Educational technology, Accessibility in education

Introduction

Education is a fundamental human right and a key driver of social mobility, economic participation, and personal development. However, approximately 16% of the global population, over 1 billion people, live with some form of disability, many of whom face

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significant educational barriers (Gvetadze et al., 2023). Despite international efforts such as the United Nations Convention on the Rights of Persons with Disabilities (CRPD) and the Sustainable Development Goal 4 (SDG4), students with disabilities still encounter multiple obstacles that hinder equitable participation in education (Hendriks, 2007). These obstacles manifest across physical, systemic, and technological dimensions. For example, in traditional learning environments, students with disabilities continue to face significant challenges that hinder their participation (Rahmi et al., 2024). Beyond physical and technological constraints, such as inaccessible spaces and non-inclusive digital platforms, systemic issues further make inclusion even more challenging by limiting accessibility, engagement, and equal learning opportunities. Examples of systemic issues include lack of individualized support, inadequate support, etc. (Ainscow, 2020). These barriers intersect and compound, creating complex obstacles that limit educational opportunities for students with disabilities.

According to the World Health Organization (WHO), disability refers to impairments, limitations in activity, and restrictions in participation that result from the dynamic interaction between an individual's health condition and contextual elements, including societal attitudes and environmental barriers (Disabilities|WHO, 2025). Over time, the definition of disability has evolved, shifting from a purely medical perspective to more inclusive and socially grounded frameworks. This paper adopts a broad and contemporary understanding of disability that explicitly includes neurodiversity, a term encompassing neurological differences such as autism, ADHD, dyslexia, and other cognitive conditions. These differences are increasingly recognized not as deficits but as natural variations in human cognition and behavior, consistent with the neurodiversity paradigm (Goldberg, 2023). This framework challenges traditional binary definitions of ability/disability and promotes inclusive practices that support varied neurological profiles within society (Silvester & Rankine, 2024). Recent scholarship further emphasizes that understanding disability in today's world requires integrating both historical context and evolving concepts that account for social, cultural, and neurological diversity (Pokhriyal, 2024). Therefore, throughout this paper, disability will be analyzed through a social and neurodiversity-informed lens, which aligns with contemporary efforts to promote dignity, accessibility, and equal opportunity for all individuals.

In the last decades, Smart Learning Environments (SLEs) have emerged as a possible solution to these challenges, providing adaptable, interactive, and technology-driven learning experiences (Cheung et al., 2021a; El-Sabagh, 2021). An SLE is defined as a hybrid learning system that personalizes learning based on students' requirements, preferences, and real-time performance data, while ensuring desired outcomes (Rosmansyah et al., 2023). SLEs leverage artificial intelligence (AI), Augmented Reality (AR) data analytics, multimodal flexibility, collaboration, and interactive platform. These components reflect the broader pillars of SLEs, namely intelligence, adaptability, collaboration, data-driven insights, and flexibility. Each of them contributes to creating a personalized and efficient learning experience. Intelligent tutoring systems, for instance, provide real-time feedback and adaptive assessments (Rosmansyah et al., 2023). Similarly, adaptive learning technologies modify instructional content based on individual learning styles and progress (Cheung et al., 2021b). Collaborative learning

platforms enhance teamwork and knowledge sharing. In parallel, smart classrooms integrate IoT and sensor-based systems to optimize learning conditions (Wang, 2021). Additionally, learning analytics and big data track student performance, providing insights to refine teaching strategies (Nurjaman, 2019). Lastly, flexible and blended learning modalities integrate digital and in-person education for greater accessibility (Faruqi & Supangkat, 2020). Collectively, these factors work to create an ecosystem that promotes adaptive instruction and inclusive participation, ultimately improving teaching effectiveness and learner success.

SLEs are becoming increasingly significant in mainstream education due to their ability to enhance learning efficiency, accessibility, and engagement through advanced technology. The rise of digital education, particularly in the wake of global disruptions such as the COVID-19 pandemic, has accelerated the integration of SLEs into traditional and online classrooms (Wu et al., 2020). Several major technology corporations and governmental bodies have launched initiatives to support the integration of SLEs in mainstream education, aiming to enhance learning efficiency, accessibility, and engagement. For example, Microsoft's Education AI Toolkit provides educators with resources and strategies to implement AI-driven learning solutions (*Microsoft 365 AI Education Tools* | Microsoft Education, n.d.). Intel has played a key role in advancing smart learning environments by integrating ICT into education, focusing on technology integration, policy recommendations, and sustainability to enhance national competitiveness and educational transformation (Price, 2015). Governments are also developing standards and frameworks to guide digital technology integration in education while promoting education for sustainable development through smart classrooms (Cebrián et al., 2020; Hoel & Mason, 2018). These initiatives help prepare learners for Industry 5.0 by incorporating AI, IoT, and robotics, fostering lifelong learning, and enhancing engagement through adaptive technologies (Adel, 2024; Cheung et al., 2021a, 2021b).

SLEs have the potential to promote inclusion by addressing the unique challenges that students with disabilities meet, like communication barriers and cognitive processing abnormalities (El-Sabagh, 2021). For instance, SLEs can incorporate differentiated instruction and universal design for learning, which are essential for accommodating the varied learning styles and challenges faced by students with disabilities (Kaimara et al., 2021; Thapliyal et al., 2021). Additionally, smart technologies such as smart toys and smart speakers have been shown to enhance learning and communication skills in children and adults with intellectual disabilities, respectively, by providing engaging, interactive, and supportive learning contexts (Ekin et al., 2018; Smith et al., 2021). These environments not only improve educational outcomes but also promote inclusivity and accessibility, thereby reducing the information divide experienced by people with disabilities (Nam & Park, 2017; Yu, 2024). Despite existing accessibility standards and laws, their practical application remains inconsistent, as seen in Portugal, where educators struggle to effectively deliver accessible content (Machado & Santos, 2023). Research on accessibility within Open Educational Resources and open educational practices is still developing (Zhang et al., 2020). Additionally, SLEs can exacerbate the information divide for people with disabilities due to limited access to smart devices and the necessary digital skills

(Nam & Park, 2017). Adaptive e-learning environments, while crucial for diverse learning needs, face implementation challenges due to educators' lack of knowledge and technological competence, highlighting the need for automated tools to enhance accessibility (Ingavélez-Guerra et al., 2022). Moreover, there is insufficient integration of assistive technologies within educational resources, limiting accessibility improvements for students with disabilities (Zhang et al., 2020). Infrastructure and policy gaps further hinder inclusivity, as seen in Qatar, where ICT infrastructure deficiencies were identified in government and semi-government institutions (Othman et al., 2023). High costs, technical barriers, and limited teacher readiness also restrict the adoption of advanced technologies like AI and VR, preventing their effective use in fostering inclusive learning environments (Chalkiadakis et al., 2024). Overall, while there are promising developments in the field of SLE, significant gaps and limitations remain, particularly in terms of accessibility and inclusion for students with disabilities.

These gaps highlight the need for a systematic synthesis of how Smart Learning Environments are being developed and applied to support students with disabilities. The next section outlines this gap and section 3 presents the guiding research questions.

Background

Inclusive education ensures that all students, including those with disabilities, have equitable access to high-quality educational opportunities (Cologon, 2022). Over the past two decades, international frameworks such as the CRPD and the SDG 4 have emphasized the need for inclusive and accessible education. These frameworks mandate the removal of systemic barriers that hinder students with disabilities from fully participating in educational settings.

However, despite these global efforts, students with disabilities confront numerous challenges, such as inaccessible resources, and a lack of specialized support. One major barrier is the lack of accessible digital and physical learning environments. Traditional educational materials and teaching methodologies do not accommodate students with sensory, motor, or cognitive disabilities (Lynch et al., 2024). Additionally, insufficient teacher training in inclusive pedagogical strategies and a lack of assistive technologies further widen the accessibility gap (Zhang et al., 2024).

Effective inclusion requires not only policy enforcement but also the integration of technological solutions that cater to diverse learning needs. In this context, smart learning environments have emerged as a transformative solution to these challenges. SLEs represent an evolution in educational technology, shifting from traditional classroom-based instruction to technology-driven, personalized, and interactive learning experiences. SLEs utilize emerging technologies to tailor educational content to the needs of individual learners, enabling real-time adaptation and feedback (García-Tudela et al., 2020).

A number of real-world implementations demonstrate SLEs' inclusive potential. For example, iTalk2Learn combines cognitive teaching with speech recognition to help struggling young students (Grawemeyer et al., 2015). Smart Classroom efforts in South Korea combine AI and sensor technology to tailor instruction, resulting in favorable

outcomes for a varied range of learners (Lim & Kye, 2019). Similarly, CAST's Universal Design for Learning (UDL) architecture has been implemented in schools across the United States to provide inclusive, adaptable digital learning environments (Rao et al., 2023).

Since SLEs cater to individual learners' unique needs, they are particularly promising for students with disabilities. These environments can include assistive technologies (e.g., screen readers, speech-to-text tools, or alternate input devices) to support students with sensory or motor disabilities. Despite their promise, many SLEs continue to fall short in addressing the unique requirements of students with disabilities due to inadequate adaptability and inconsistent accessibility standards.

Theoretical Framework This study adopts the Ecology of Inclusive Education in Innovative Learning Environments (ILEs) model (Page et al., 2023) as its guiding framework. The model views inclusion as an interaction between design, pedagogy, practice, and systemic supports, which together create equitable opportunities for students with disabilities.

Applied to SLEs, this framework highlights that:

- Students with sensory and physical disabilities rely strongly on design affordances such as multimodal tools and accessible interfaces.
- Students with cognitive and neurodevelopmental disabilities benefit from individual and academic affordances such as adaptive scaffolding, gamification, and personalized AI support.
- Sustainable inclusion also requires systemic affordances, including educator training, policy frameworks, and institutional leadership.

By using this ecological perspective, the study frames inclusive SLEs not only as technological systems but as learning ecosystems where tools, pedagogy, and institutional contexts must work together to support all disability groups.

Research Gap and Objectives While SLEs present numerous opportunities for enhancing inclusive education, they also face significant challenges in accessibility and implementation. Most research focuses on their benefits for the general student population, often overlooking the specific needs of students with disabilities (Zhang et al., 2024). Although SLEs have been widely studied, existing review studies have explored various aspects of technology-enhanced learning for students with disabilities, but few systematically examine how SLEs specifically support students with disabilities.

Over the past decade, several literature reviews have explored aspects of inclusive education and educational technologies, but most focus on general assistive tools, specific technologies (e.g., AR, VR), or isolated education levels, rather than holistically examining SLEs. For example, Lynch et al. (2024) explored educational technology in primary education, identifying accessibility gaps but without a specific focus on SLEs. Another review has focused on the role of assistive technologies in higher education for students with learning difficulties but lacks the evaluation of how SLEs integrate these technologies comprehensively (Yenduri et al., 2023). In another review conducted by (Cai et al., 2021), the authors focused on AR in accessible learning, noting its potential but the lack of investigation of their

long-term accessibility for learners with disabilities. Additionally, Mukhtarkyzy et al. (2025) conducted a review examining the impact of educational technologies on learning outcomes among diverse student populations, with a particular focus on assistive technology interventions for students with disabilities. Their findings highlight the potential of assistive technologies while underscoring the need for stronger empirical evidence and more refined, user-centered design approaches. Similarly, Araújo & Andrade (2021) examined programming education for students with cognitive disabilities, revealing limited but growing interest in accessible coding tools and highlighting areas for future research. These studies highlight a growing interest in inclusive digital education, yet underscore a significant gap: there is no comprehensive systematic literature review that evaluates how SLEs are developed and applied to support students with disabilities across diverse educational contexts. Furthermore, most existing reviews do not assess the full integration of inclusive frameworks such as Universal Design for Learning (UDL) and assistive technologies within SLEs.

This gap is particularly critical as digital education becomes increasingly embedded in both formal and informal learning environments. Without a structured synthesis of evidence on the inclusive potential of SLEs, educators, policymakers, and technology developers are limited in their ability to design truly equitable and effective learning environments. As SLE technologies evolve and global emphasis on inclusive education increases, a comprehensive review is essential to consolidate current knowledge, expose systemic accessibility barriers, and provide actionable, evidence-based recommendations.

Research Questions To address this gap, the objective of this systematic literature review is to analyze and synthesize current knowledge on SLEs for students with disabilities by identifying their applications, challenges, and best practices. Specifically, this review is guided by the following research questions:

RQ1. What are the trends in the existing literature on inclusive smart learning environments for people with disabilities in terms of publication year, publication type, and venue?

RQ2. What is the distribution of studies in terms of disability types and educational levels targeted?

RQ3. What are the research methodologies and the current applications of SLE that specifically target the empowerment of students with disabilities?

RQ4. What are the key challenges and gaps faced by students with disabilities when engaging with SLEs?

RQ5. What are the best practices for implementing SLEs that promote equity, inclusivity, and empowerment in diverse educational contexts?

By answering these questions, this review contributes a comprehensive, evidence-based foundation to guide educators, designers, and policymakers in advancing truly inclusive SLEs. This SLR synthesizes findings from studies published between 2019 and 2024 across six major academic databases: Scopus, IEEE Xplore, ACM Digital Library, and Google Scholar. By rigorously reviewing 43 studies that match the inclusion criteria, this SLR gives a complete overview of how SLEs are used to support students with disabilities. It exposes existing systemic barriers while also

making actionable ideas for educators, policymakers, and technology developers to design and implement SLEs that promote inclusivity and empowerment in diverse educational contexts.

The paper is structured as follows: Section 1 introduces the topic and presents the context and rationale behind this review. Section 2 presents the background and significance of inclusive smart learning, and outlines the research gap and study objectives. Section 3 details the methodology used for the systematic review. Section 4 presents and analyzes findings organized by the research questions. Section 5 discusses the implications of the findings. Section 6 concludes the paper and suggests future research directions.

Research methodology

This research employed a systematic literature review approach to deliver unbiased, comprehensive, and replicable insights in a specific research domain (Kitchenham, 2004). The study highlights its transparency by outlining the search and selection processes and providing a numerical overview of the articles obtained. Nonetheless, such reviews can encounter issues stemming from technical constraints, especially those linked to database searching (Boell & Cecez-Kecmanovic, 2010).

Search strategy

The search strategy was executed across six leading databases: Scopus, ACM Digital Library, ScienceDirect, IEEEExplore, SpringerLink, and Google Scholar. The search query was designed to target titles, abstracts, and indexed terms. The selection criteria prioritized peer-reviewed journal articles, conference papers, and book chapters that had been accepted or published. To construct a comprehensive and inclusive search query, keywords were derived from the principal themes of the research questions and included alternative spellings, synonyms, and acronyms. Boolean operators were employed to structure the query: “OR” was used to connect similar terms within the same category, while “AND” was applied to link terms across different categories. The final query was constructed as follows:

“student” OR “students” OR “learner” OR “learners”) AND (“disability” OR “disabilities” OR “students with disabilities” OR “learners with disabilities” OR “disorder” OR “disorders” OR “impairment” OR “difficulties”) AND (“smart learning environment” OR “smart learning environments” OR “intelligent learning environment” OR “intelligent learning environments” OR “SLE” OR “inclusive education”).

This structured approach ensured a systematic and thorough retrieval of studies initially identified 2297 unique papers for the first round of inclusion/exclusion screening.

Inclusion and exclusion criteria

To ensure the inclusion of relevant studies, a two-stage screening process was implemented. Initially, 70 duplicate entries were removed. In the first phase, titles and abstracts were evaluated, followed by a detailed review of full texts to verify alignment with the research questions. The scope of this review was limited to studies published between 2019 and 2024. This time frame was selected based on

the notable acceleration in Smart Learning Environment (SLE) research in recent years. While foundational discussions began earlier, it is during this five-year window that the field matured in terms of both conceptual clarity and real-world implementation. Recent bibliometric analyses highlight this period as a key phase of growth, characterized by increased focus on intelligent technologies, inclusion strategies, and educational accessibility within SLEs (Maulidiya et al., 2024). The selected time frame ensures that the review reflects the most current developments and insights.

Studies were eligible for inclusion if they (1) described a smart learning environment involving individuals with disabilities, (2) were published between 2019 and 2024, and (3) were written in English. During this initial phase, the first two authors independently reviewed the abstracts to identify relevant studies. Specifically, 1114 abstracts were reviewed by the first author and 1113 by the second author. This process resulted in the selection of 245 studies, while 1982 publications were excluded due to reasons such as being outside the scope of this review ($n = 1882$) or not including individuals with disabilities ($n = 26$). Additionally, 68 systematic literature reviews, one thesis, and five publications of unrelated types (e.g., short papers and extended abstracts) were excluded.

In the second stage, the same two authors conducted a comprehensive review of the remaining papers by applying the inclusion and exclusion criteria and extracting data relevant to the research questions. Of the 245 papers, 14 exhibited disagreements regarding inclusion between the first two authors; these were subsequently reviewed by a third author to resolve discrepancies and extract the necessary data. This phase led to the exclusion of an additional 202 papers for similar reasons, including being outside the review's scope ($n = 146$) or lacking inclusion of individuals with disabilities ($n = 28$), as well as the exclusion of 17 systematic literature reviews, three non-English papers, and eight publications of unrelated types (e.g., short papers and extended abstracts). Ultimately, 43 papers met the inclusion criteria and were retained for detailed analysis in the final corpus. The entire selection process is visually summarized in Fig. 1.

Data extraction and analysis

The initial two authors conducted a comprehensive review of the corpus included in this Systematic Literature Review. Bi-weekly meetings were convened to discuss and refine the data extraction process, ensuring its alignment with the research questions. Through iterative discussions and refinements, a standardized reporting form was developed. This form captured: (1) study-specific details, including the study's aim and participant information (age group, diagnosis, number of participants, and gender); (2) methodological aspects, detailing the data collection and analysis methods employed; (3) study outcomes, specifying the type of prototype (e.g., serious game, tangible tool, framework) and its target environment; and (4) the challenges and recommendations identified in the included studies.

Each author independently extracted data using the standardized form, and subsequently, both authors cross-checked each other's extraction results to ensure data accuracy. In instances where information was unavailable, the coders

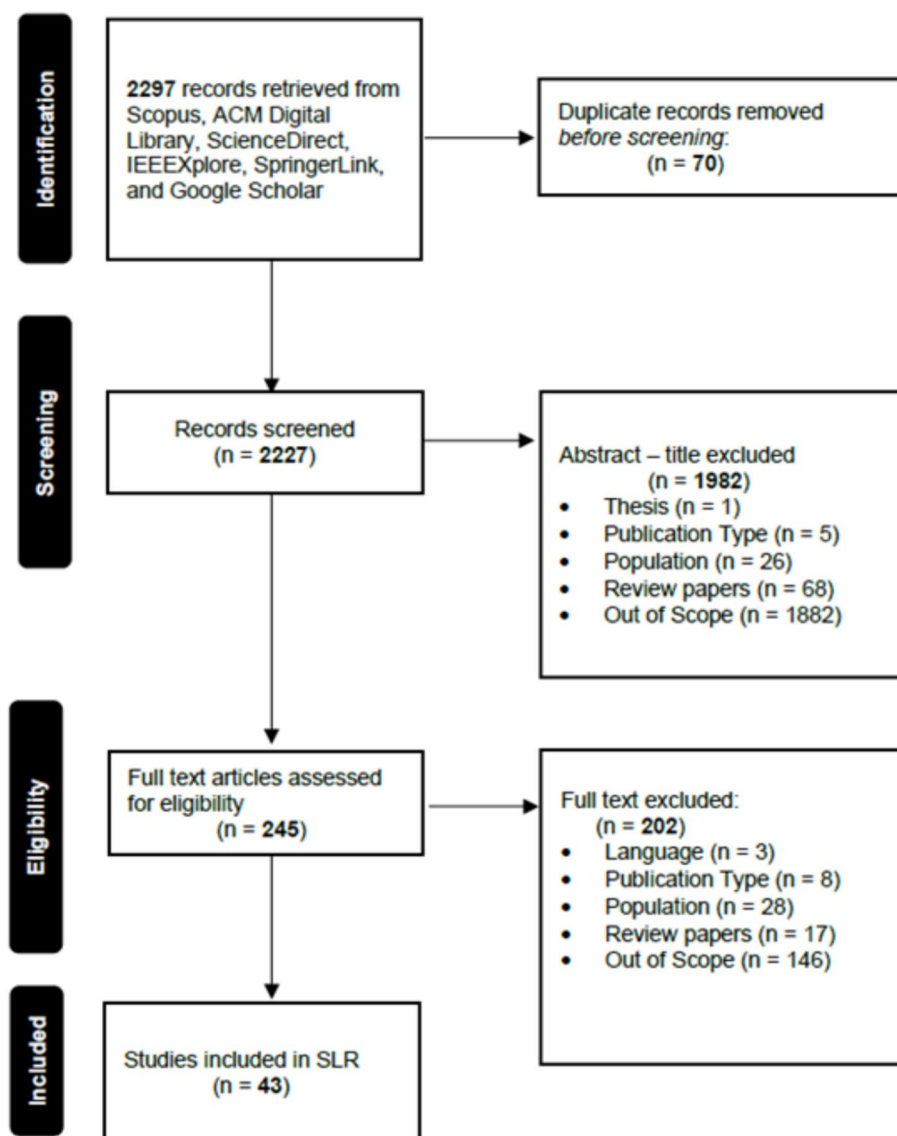


Fig. 1 Systematic literature selection process following PRISMA framework

appropriately recorded “N/A” or “not mentioned.” Finally, all extracted data were consolidated into an Excel spreadsheet, thereby creating a unified and organized dataset for analysis.

While this review did not include primary data collection from domain experts, the integration of expert validation is increasingly recognized as an important step in enhancing the relevance and rigor of systematic literature reviews, particularly in educational technology and inclusive design (Ngadengon et al., 2024). Future iterations of this study may employ Delphi methods or expert panels to assess the implications of findings and validate identified trends and frameworks.

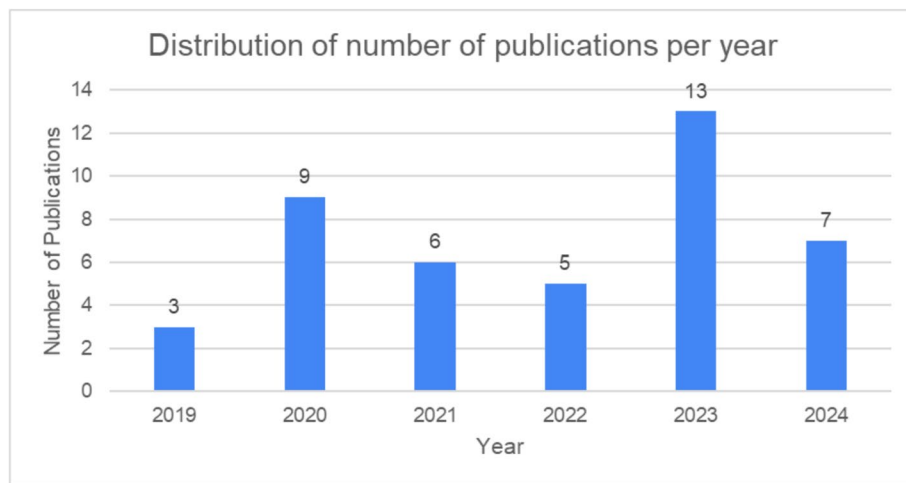


Fig. 2 Distribution of selected papers by year of publication

Table 1 Prominent publication venues (two or more times listed in the corpus)

Publication type	Name	#
Conference	CHI	4
Journal	ACM Transactions on Accessible Computing	4
Journal	Teaching and Teacher Education	3
Journal	Heliyon	2
Journal	IEEE Transactions on Learning Technologies	2
Journal	ASSETS	2

Results

This section summarizes the results of this SLR paper in response to the research questions defined previously in section "[Background](#)".

Trends in the Literature on Inclusive SLEs (RQ1)

After a thorough review, 43 peer-reviewed studies were identified as relevant to this study. These studies, which explored the inclusion of students with disabilities through smart learning environments, exhibited variation in publication trends over the past five years. As shown in Fig. 2, only 3 papers were included in 2019. This number increased in 2020 to 9 papers, indicating growing interest in the field. However, the subsequent years saw slight fluctuations: 6 papers in 2021 and a further decrease to 5 papers in 2022. The most notable surge occurred in 2023, with 13 papers published, suggesting a peak in research activity during that year. In 2024, and until August 2024, the number stabilized at 7 publications. Overall, the trend reveals initial growth, a period of moderate fluctuation, and a significant peak in 2023 before returning to earlier levels.

Table 1 summarizes the publications that appeared at least twice in the final corpus. Most of the papers included were conference articles ($n=26$) of which 17 were from ACM conferences and nine from IEEE conferences. On the other hand,

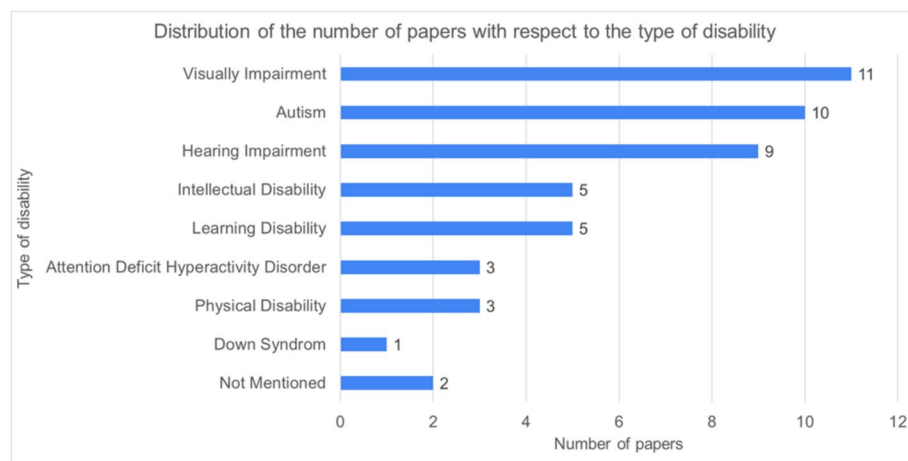


Fig. 3 Distribution of the number of papers with respect to the type of disability

17 papers were from journals in which Elsevier had the majority ($n=9$), followed by ACM ($n=4$), IEEE ($n=2$), and Emrad and MDPI in which each had only one included publication.

Distribution of Studies by Disability Type and Educational Level (RQ2)

Participants

The participant demographics across the reviewed studies reflect a broad spectrum of educational contexts and types of disabilities. As illustrated in Fig. 3, the distribution of papers reveals a predominant research focus on visual impairment, Autism Spectrum Disorder (ASD), and hearing impairment, while also addressing other categories such as learning disabilities, physical disabilities, and intellectual and developmental disabilities (e.g., Herrera Poma et al., 2023; Patiño-Toro et al., 2023; Sulek et al., 2019)). This varied distribution underscores the multifaceted nature of inclusive education research and highlights both well-explored areas and emerging fields that warrant further investigation. By analyzing diverse participant profiles from educators and specialists to students across various age groups, this section examines how demographic diversity affects the design and implementation of inclusive educational interventions, providing important insights into their impact.

Autism spectrum disorder Ten studies focused on supporting children with ASD. For example, Sulek et al., (2019) engaged 21 participants (including parents, teachers, and special education staff) to examine an online tool supporting school transitions. Similarly, Van Der Steen et al. (2020) involved 43 specialists to assess the professional needs in teaching children with ASD. Single-case investigations such as Gilabert-Cerdá et al. (2023) and Amaral et al. (2021) detailed the impact of AR and alternative communication platforms on individual students with ASD. Additionally, studies by Almourad and Bataineh (2020), Sobel et al. (2024), Garzotto et al. (2020), and Dinét et al. (2022) evaluated interventions ranging from eyetracking-analyses to hybrid robotics systems, with sample sizes varying from single digits to 65 children. Notably, Piki and Markou (2023)

addressed co-occurring conditions by examining the role of digital games for students with both ADHD and ASD, while Larios and Zetlin (2023) evaluated professional development for teachers working with students who present a combination of ASD, physical impairments, and intellectual disabilities. These findings correspond to individual and social affordances in the ecological framework (Page et al., 2023), where structured routines and gamified supports enhance self-regulation while collaborative tools foster social participation.

Hearing impairment Seven studies investigations targeted learners with hearing related-challenges. The studies of Patiño-Toro et al. (2023), Sun et al. (2022), Elmahmoudi et al. (2021), Xohua-Chacon et al. (2019), Ayu Anggraini et al. (2021), and Co et al. (2020) focused on developing inclusive methodologies and technological tools for individuals with hearing impairment. These studies included a varied range of individuals, from small-scale pilot groups (e.g., 2–5 individuals in Xohua-Chacon et al. (2019) and Pal et al. (2023)) to larger samples, such as 50 participants in Ayu Anggraini et al. (2021). Notably, Pal et al. (2023) examined the integration of realtime—closed captioning and live transcription for DHH students, while Co et al. (2020) evaluated an elearning—application designed for deaf and mute students learning Filipino Sign Language.

Learning disability Research addressing learning disabilities, such as dyslexia and processing difficulties, was represented in studies by Bressane et al. (2024), Al Omoush et al. (2023), Fung et al. (2023), Popov and Ivanova (2020), and Zammouri et al. (2024). These studies employed diverse methodologies, including AI based decision support systems (Bressane et al., 2024), gamified mobile applications (Fung et al., 2023), and machine learning based pedagogical agents (Al Omoush et al., 2023). Participants ranged from large cohorts such as the 822 engineering undergraduates in Bressane et al. (2024) to smaller groups in more focused evaluations (Popov and Ivanova (2020); Zammouri et al. (2024)). Collectively, these studies underscore the potential of adaptive and personalized educational technologies in supporting students with learning disabilities. This emphasis on AI-based scaffolding and adaptive technologies reflects the academic affordances identified in our framework, extending cognitive support and enabling flexible access to content.

Physical disability Investigations focusing on physical disabilities were relatively few but provided valuable insights into assistive strategies for mobility or neuromotor challenges. AlTaleb et al. (2024) explored the educational experiences of 20 physically disabled female students in higher education, while Canet and Tonidandel (2022) reported on case studies with 2 elementary school students with severe neuromotor disabilities. These studies highlight the specialized adaptations required to support physical disability in educational robotics and other learning environments. The use of robotics and adaptive devices highlights physical and digital design affordances, where inclusive SLEs reduce interaction barriers through accessible interfaces.

Intellectual disability A subset of the research concentrated on intellectual and developmental disabilities, including Down syndrome. Cunha-Pérez et al. (2024) evaluated adaptive instructional strategies for 20 students with Down Syndrome, and studies by Kai-

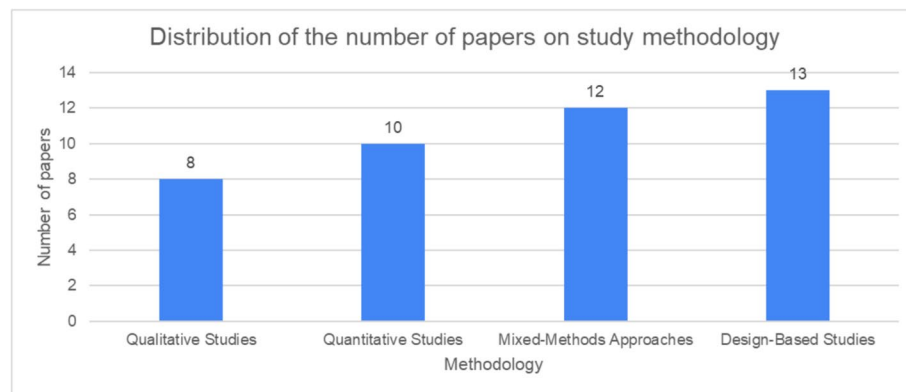


Fig. 4 Distribution of the number of papers on the study methodology

mara et al. (2021), Venkatasubramanian et al., (2023), and Howard et al. (2021) involved participants identified as having intellectual disabilities. Although sample sizes in these investigations varied from focus group discussions to controlled experimental designs, the findings collectively emphasize the importance of tailored, adaptive learning environments for this population. These studies demonstrate individual affordances through adaptive instructional strategies, though the relative absence of systemic affordances, such as teacher training or institutional supports, remains notable within our framework.

Visual impairment Visual impairment was one of the most frequently addressed categories in the reviewed literature. Herrera Poma et al. (2023) focused on a robotic device for teaching Braille to visually impaired children aged 6–12, while Lobo et al. (2020) and Rocha et al. (2023) engaged participants ranging from children to adults in workshops and collaborative activities. Further, Neto et al. (2023) and Kearney-Volpe et al. (2023) detailed workshops designed for blind participants using screen readers, and Metatla et al. (2020) provided insights into interactive multisensory environments. Additional contributions by Neto et al. (2021), Kearney-Volpe et al. (2021), Neto et al. (2020), and Shaheen (2024) expanded on these approaches, with several studies also including mixed ability settings to promote inclusive interactions between visually impaired and sighted learners. Braille robotics, screen readers, and multisensory tools illustrate individual and academic affordances in the ecological model, ensuring literacy and participation through multimodal access pathways.

Mixed and unspecified disability A few studies, including those by Khaksar et al. (2019), Passmore et al. (2023), and Gutiérrez Y Restrepo and Floris (2022), did not specify a particular disability type but nonetheless contributed to the broader discourse on inclusive educational technologies. Moreover, some studies reported participants with multiple or co-occurring disabilities such as Kukushkina et al. (2022), which involved hearing impairment, visual impairment, ADHD, and intellectual disabilities highlighting the complex, overlapping challenges encountered in inclusive settings.

Current Research methodologies in the selected studies (RQ3)

Current Research Methodologies of SLE in the Selected Studies The reviewed studies employed a variety of research methodologies to design, test, and evaluate SLE solutions aimed at empowering students with disabilities. Four main methodological approaches emerged.

As illustrated in Fig. 4, the methodologies used in the included studies fall into four main categories: (1) Qualitative Studies, (2) Quantitative Studies, (3) Mixed-Methods Approaches, and (4) Design-Based Studies. Design-based studies constitute the largest share of literature, followed closely by mixed-methods research. Quantitative studies are also well-represented, whereas qualitative studies form the smallest category.

Qualitative studies Studies in this category predominantly rely on qualitative methods to gain deep insights into the experiences, perceptions, and contextual factors affecting inclusive education. These investigations use techniques such as in-depth interviews, focus groups, autoethnographies, and observational case studies. For example, Sulek et al. (2019) employed qualitative interviews with 21 stakeholders including parents, teachers, and special education staff to assess the social validity of an online tool designed to support transitions to primary school for children with autism. Similarly, AlTaleb et al. (2024) conducted semi-structured interviews with physically disabled female students in higher education, using thematic analysis to uncover the facilitators and barriers these students face, thereby highlighting the nuanced challenges within inclusive academic settings. In Amaral et al. (2021), a detailed case study of a 9-year-old child with ASD was carried out to evaluate the impact of the Expressia platform on communication and cognitive development, providing rich, contextual insights despite its single-case design. Furthermore, Jain et al. (2020) used autoethnographic methods to document the lived experiences of disabled graduate students, thereby offering a reflective account of how disability accommodations affect academic and social experiences over time. Collectively, these qualitative studies contribute profound insights that inform the development of context-sensitive interventions, as they illuminate the complex realities of inclusive education environments (see also Van Der Steen et al. (2020); Xohua-Chacon et al. (2019)).

Quantitative studies Quantitative and experimental studies in the review are characterized by their systematic use of statistical analyses, controlled conditions, and measurement instruments to test hypotheses and evaluate intervention outcomes. These studies commonly adopt designs such as pretest-post-test evaluations, quasi-experimental methods, or correlational studies. For instance, Sun et al. (2022) compared the effects of AR with traditional pictorial methods on preschool children's comprehension of sequential time, providing empirical evidence that AR can enhance cognitive understanding, particularly among hearing-impaired children. Gilabert-Cerdá et al. (2023) offer another example by utilizing a quasi-experimental, single-participant design to measure improvements in different types of attention in a child with ASD, with data collected over 10 intervention sessions. In Cunha-Pérez et al. (2024), a controlled experiment involving students with Down Syndrome demonstrated that adaptive instructional strategies in a computer-based learning platform significantly improved second-language acquisition, as measured by pre- and posttest assessments. Additionally, Ayu Anggraini et al. (2021)

used a correlational design to establish a significant positive relationship between the intention to use Web Captioner and academic self-efficacy among deaf students, thereby highlighting the potential of assistive technologies in bridging communication gaps. Co et al. (2020), Dinet et al. (2022), and Zammouri et al. (2024) further expand the quantitative evidence base by measuring behavioral changes, learning outcomes, and cognitive load through rigorous experimental protocols.

Collectively, these studies provide statistically robust data that support evidence-based practices and demonstrate the efficacy of technological interventions in diverse educational contexts.

Mixed methods approaches Mixed-methods studies integrate both qualitative and quantitative techniques to offer a comprehensive perspective that captures measurable outcomes alongside the contextual factors influencing them. This integrative approach allows researchers to triangulate data, thereby enriching the interpretation of complex phenomena.

For example, Larios and Zetlin (2023) combined surveys, focus group discussions, and interviews to assess the impact of professional development programs on teachers' abilities to implement inclusive education practices. The study revealed both statistically significant improvements and detailed narratives of practical challenges. Similarly, in Fung et al. (2023), a mixed-methods design was used to evaluate a gamified mobile application for students with dyslexia, where quantitative handwriting scores and time analysis were complemented by qualitative feedback from the participants. Rocha et al. (2023) adopted a similar approach by employing both quantitative measures (e.g., completion times and engagement scores) and qualitative coding of interactions during collaborative coding activities involving children with mixed visual abilities. In addition, Kearney-Volpe et al. (2023) evaluated a remote web development workshop for blind participants through pre- and post-workshop surveys, quizzes, and in-depth interviews, providing insights into both the effectiveness of the training and the experiential aspects of learning with assistive technology. Garzotto et al. (2020) and Shaheen (2024) also demonstrate the value of mixed methods in capturing both the behavioral and emotional dimensions of learners' interactions with technology. Overall, mixed-methods research in this field successfully bridges numerical data with rich descriptive information, thereby offering a holistic understanding of how interventions are performed in real-world educational settings.

Design-based studies Design-based and iterative prototyping studies focus on the creation, development, and refinement of educational tools and technological interventions, emphasizing user-centered design and continuous improvement. Researchers in this category engage in iterative cycles of prototyping, testing, and refinement to ensure that the developed tools meet the needs of diverse learner populations. For instance, Patiño-Toro et al. (2023) outline a comprehensive methodology for designing Massive Open Online Courses (MOOCs) tailored to Deaf and hard-of-hearing individuals, combining literature review insights with expert consultations to inform the design process. Al Omoush et al. (2023) describe the development and testing of a machine learning-based pedagogical agent that integrates visual and auditory modalities to assist primary school students,

with particular attention to accessibility for those with sensory disabilities. Similarly, Neto et al. (2023) detail an iterative design process for a mediator robot aimed at fostering inclusive conversations among children with and without visual impairments, while Howard et al. (2021) report on the co-design and qualitative evaluation of a mobile app intended to educate individuals with intellectual and developmental disabilities about abuse recognition. Other studies such as Elmahmoudi et al. (2021), Gutiérrez Y Restrepo and Floris (2022), Herrera Poma et al. (2023), Kaimara et al. (2021), Lobo et al. (2020), Metatla et al. (2020), Neto et al. (2021), Popov and Ivanova (2020), Sobel et al. (2024), and Venkatasubramanian et al. (2023), further illustrate the application of design-based research in education. These studies have contributed to the development of various innovative tools, including smart classrooms, tactile coding kits, and robotics for teaching Braille. Also, they frequently incorporate formative feedback, usability testing, and iterative adjustments, underscoring the critical role of design-based research in advancing adaptive and accessible educational technologies.

Applications of SLE in the Selected Studies

Smart learning environments integrate a wide range of tools and technologies to enhance accessibility, engagement, and inclusivity across diverse educational contexts. In this section, we provide an overview of the various technological solutions implemented in SLEs as reported by the reviewed studies. The discussion is organized into three main sub-sections: (I) K-12 and Primary/Elementary School Tools and Technologies, (II) Higher Education Tools and Technologies, and (III) eLearning- and Mobile Applications. Each sub-section highlights innovations tailored to the specific needs of its target user group, ranging from prototype online tools and augmented reality applications in early education (e.g., Gilabert-Cerdá et al., 2023; Sulek et al., 2019; Sun et al., 2022) to IoT-enabled smart classrooms and integrated higher education platforms (e.g., Elmahmoudi et al., 2021; Gutiérrez Y Restrepo & Floris, 2022; Kukushkina et al., 2022). Moreover, the section examines adaptive eLearning—systems and mobile applications designed to support learners with diverse challenges, including those with sensory, intellectual, and developmental disabilities (e.g., Co et al., 2020; Cunha-Pérez et al., 2024; Patiño-Toro et al., 2023; Popov & Ivanova, 2020). Through this comprehensive synthesis, the section underscores the multifaceted nature of SLEs and their critical role in fostering inclusive educational experiences.

K-12 and primary/elementary school

This sub-section focuses on the diverse range of technologies applied in K-12 and primary/elementary school environments to foster inclusivity. Table 2 provides a summary of tools and technologies in K-12 and primary/elementary school environments. Several studies have developed prototype systems and interactive devices aimed at addressing the varied needs of young learners. For instance, an online tool was designed to support transitions to primary school for children with autism by connecting teachers, parents, and support staff (Sulek et al., 2019). AR has been employed in preschool and school settings to enhance the comprehension of abstract concepts, especially benefiting hearing-impaired-children (Gilabert-Cerdá et al., 2023; Sun et al., 2022). In addition, various robotic systems have been developed to support skill acquisition and social interactions.

Table 2 Summary of tools and technologies in K-12 and primary/elementary school environments

Paper	Target environment	Type of solution
Sulek et al. (2019)	Primary school transitions	Prototype online tool
Sun et al. (2022)	Kindergartens, rehabilitation centers	AR educational tool
Gilabert-Cerdá et al. (2023)	School	AR application for attention enhancement
Canet and Tonidandel, (2022)	Elementary school	Assistive robotics with adaptive switches
Pal et al. (2023)	Face-to-face classrooms	Real-time captioning prototype
Xohua-Chacon et al. (2019)	Classroom	Tangible user interface for learning relational algebra
Herrera Poma et al. (2023)	Classroom and home	Robotic teaching aid with IoT capabilities
Rocha et al. (2023)	Classroom	Hands-on tangible robotics for coding and spatial learning
Neto et al. (2023)	School	Social robot for mediating group interactions
Sobel et al. (2024)	Kindergarten	Tablet application fostering inclusive play
Garzotto et al. (2020)	Primary schools	Interactive Multisensory Environment (iMSE)
Ayu Anggraini et al. (2021)	Classroom	Assistive communication tool (real-time captioning)
Neto et al. (2020)	Classroom	Haptic feedback robotic system
Shaheen (2024)	K-12 (across schools)	Framework for accessible instructional tools
Dinet et al. (2022)	School	Robotics system to support joint attention and peer interaction

Educational robotics have been used to empower students with severe neuromotor disabilities (Canet & Tonidandel, 2022) and to teach Braille to visually impaired children (Herrera Poma et al., 2023). Other innovative solutions include tangible coding kits that encourage collaborative learning (Rocha et al., 2023), mediator robots that foster inclusive group conversations (Neto et al., 2023), tablet applications for joint media engagement (Sobel et al., 2024), and immersive environments such as the Magic Room (an Interactive Multisensory-Environment) for primary school children (Garzotto et al., 2020). Furthermore, assistive technologies such as real-time-captioning systems (Ayu Anggraini et al., 2021; Pal et al., 2023), tangible user interfaces like TanQuery for students with hypoacusis (Xohua-Chacon et al., 2019), and alternative communication platforms for children with autism spectrum disorder (Amaral et al., 2021) underscore

Table 3 Summary of tools and technologies in higher education environments

Paper	Short title/focus	Target environment	Type of solution
Elmahmoudi et al. (2021)	Smart Classroom Prototype using IoT	Engineering schools (Morocco)	IoT based smart classroom architecture
Kukushkina et al. (2022)	Inclusive Information and Educational Environment	University	Conceptual model integrating assistive technologies
Gutiérrez Y Restrepo and Floris (2022)	YachaY System for Inclusive Higher Education	Higher education institutions (Latin America)	Integrated platform with e-profiles, portfolios, and recommender systems
Jain et al. (2020)	Autoethnographic Study on Graduate School Accommodations	Graduate school (University)	Theoretical insights and recommendations (qualitative methodology)

the potential of technology to address a broad spectrum of challenges (Table 2). These tools not only improve accessibility and engagement in traditional classrooms but also provide empirical evidence of improved social interaction and learning outcomes (Dinet et al., 2022; Neto et al., 2020).

Higher education

In higher education settings, SLE tools are increasingly being designed to create accessible and inclusive learning environments that extend beyond conventional classrooms. Table 3 presents a summary of tools and technologies in higher education environments. For example, a smart classroom prototype that leverages Internet of Things (IoT) technologies has been developed to support Deaf-mute students in Moroccan engineering schools, thereby enhancing the overall learning experience for students with disabilities (Elmahmoudi et al., 2021). Another study proposed a conceptual model for an inclusive information and educational environment within universities, integrating assistive devices and adaptive digital resources to support diverse learners (Kukushkina et al., 2022). In Latin America, the YachaY system represents an integrated platform aimed at providing flexible and personalized educational opportunities for underserved populations (Gutiérrez Y Restrepo & Floris, 2022). Furthermore, an autoethnographic study provided theoretical insights into the challenges faced by disabled graduate students in navigating university accommodations, offering recommendations for improving SLEs at the higher education level (Jain et al., 2020). Together, these studies underscore the importance of innovative technological frameworks in transforming higher education into more inclusive and supportive environments (Table 3).

E-Learning and mobile applications

E-learning platforms and mobile applications represent a rapidly growing category within SLEs, providing versatile and scalable solutions for diverse learner populations. In this domain, Massive Open Online Courses (MOOCs) have been specifically tailored to meet the needs of Deaf and hard-of-hearing individuals by incorporating sign language integration, collaborative tools, and gamification techniques (Patiño-Toro et al., 2023). Adaptive -computer-based platforms have been developed to support second language learning for students with Down Syndrome (Cunha-Pérez et al., 2024) and to create personalized learning environments for dyslexic learners using semantic web technologies and -ontology-based models (Popov & Ivanova, 2020). Additionally, accessible -eLearning programs designed for visually impaired students employ advanced tools such as the iNetSim simulator and augmented reality devices like Microsoft HoloLens (Lobo et al., 2020). Mobile applications also play a significant role in this category: two separate studies detail the design and evaluation of apps aimed at empowering individuals with intellectual and developmental disabilities (I/DD) to recognize and report abuse (Howard et al., 2021; Venkatasubramanian et al., 2023). Furthermore, remote web development workshops for screen reader users have been implemented to enhance digital skills and self efficacy among blind participants (Kearney-Volpe et al., 2021, 2023), and the TEXT2FSL mobile app has been developed to facilitate Filipino Sign Language learning for deaf and mute students (Co et al., 2020). Lastly, an innovative adaptive learning environment that integrates

Table 4 Summary of tools and technologies in e-learning and mobile applications

Paper	Short title/focus	Target environment	Type of solution
Patiño-Toro et al. (2023)	MOOCs for the Deaf Community	Online learning environments	Tailored MOOCs with sign language integration and gamification
Bressane et al. (2024)	AI based Decision Support for Learning Disabilities	eLearning settings	Decision support system integrating ANN and fuzzy logic
Cunha-Pérez et al. (2024)	"Let's Speak English" Platform for DS	Classroom/eLearning	Computer based instructional platform with adaptive strategies
Popov and Ivanova, (2020)	Adaptive eLearning Framework for Dyslexic Learners	eLearning	Ontology based adaptive eLearning framework with agent based systems
Lobo et al. (2020)	eLearning for Students with Visual Impairments	eLearning	Accessible eLearning-program with tools like iNetSim and AR devices
Venkatasubramanian et al. (2023)	Mobile App for Abuse Prevention in I/DD	eLearning (mobile)	Mobile application combining slides, videos, and skill building activities
Howard et al., (2021)	Mobile App for Abuse Recognition in I/DD	eLearning (mobile)	Co-designed mobile app prototype for abuse recognition
Kearney-Volpe et al. (2023)	Remote Web Development Workshop for Screen Reader Users	eLearning	Remote workshop with synchronous lectures, tactile diagrams, and forums
Kearney-Volpe et al., (2021)	Remote Web Development Workshop (Accessible Techniques)	eLearning	Remote training workshop using accessible teaching tools and interactive methods
Co et al. (2020)	TEXT2FSL: Filipino Sign Language Translator Tool	eLearning (mobile app)	eLearning mobile application with CAI features for sign language learning
Zammouri et al. (2024)	Adaptive Learning Environment using EEG	Computer supported learning systems	Architecture integrating EEG based cognitive load estimation with adaptive content delivery

EEG based cognitive load measurements with machine learning techniques represents a-cutting-edge-approach to personalized learning (Zammouri et al., 2024). Collectively, these eLearning- and mobile solutions demonstrate how technology can provide equitable access to education by overcoming physical and cognitive barriers. Table 4 provides a summary of tools and technologies in e-learning and mobile applications.

Key Challenges and Gaps in SLEs implementation (RQ4)

The implementation of Smart Learning Environments (SLEs) for students with disabilities presents multifaceted challenges that span technical, educational, and social dimensions. This section analyzes these challenges through a systematic examination of literature.

Infrastructure and technical limitations

The foundation of successful SLE implementation rests heavily on technological infrastructure, where significant limitations persist across educational institutions. Many schools struggle with fundamental resource constraints, including limited hardware resources, outdated computers, and insufficient tablets (Kaimara et al., 2021; Metatla et al., 2020; Piki & Markou, 2023). These basic infrastructure challenges are compounded by more complex technical issues, such as system delays during IoT-based data transfer (Herrera Poma et al., 2023), difficulties with device compatibility and other technical limitations in remote settings (Howard et al., 2021; Venkatasubramanian et al., 2023), and problems with screen sharing and remote learning platforms (Kearney-Volpe et al., 2021; Rocha et al., 2023). Findings indicate that recognizing letters in varying lighting conditions and the need for enhanced hardware, such as high-quality microphones and cameras, present additional technical challenges in noisy environments (Al Omoush et al., 2023).

Educational resource management and standardization

The organizational infrastructure of educational resources in SLE implementation presents systemic challenges, particularly regarding standardization. The fragmentation of pedagogical materials and the absence of a unified curriculum are especially pronounced in game-based learning environments (Piki & Markou, 2023). Barriers to inter-institutional resource sharing and constrained access to evidence-based pedagogical materials further impede the efficacy of implementation (Sulek et al., 2019). Additionally, inconsistencies in the application of high-leverage practices and terminological disparities among educators contribute to variability in instructional delivery (Passmore et al., 2023). Findings suggest that the availability of specialized software resources remains scarce (Canet & Tonidandel, 2022), and limited access to accessible programming education materials persists (Kearney-Volpe et al., 2023).

Accessibility and usability challenges

The accessibility framework within SLEs presents multifaceted barriers that significantly affect educational outcomes. Usability challenges range from fundamental interface navigation difficulties and audio control limitations to more complex accessibility concerns, such as deficiencies in captioning and real-time transcription technologies (Pal et al., 2023; Fung et al. 2023). Learners with visual impairments encounter substantial difficulties in verifying visual elements and interpreting programmatic outputs (Kearney-Volpe et al., 2021, 2023). Similarly, hearing-impaired learners face constraints within MOOCs, alongside accessibility disparities across diverse sign language systems (Patiño-Toro et al., 2023). Reading text-based instructions was difficult for some students due to reliance on visuals (Co et al., 2020). Moreover, studies have reported that hearing-impaired students struggle with conceptualizing abstract temporal constructs, indicating specific accessibility barriers in instructional design (Sun et al., 2022).

Professional development and support

The success of SLE implementation heavily depends on educator preparedness and institutional support systems. A critical gap exists in the availability of trained instructors during early implementation phases (Lobo et al., 2020) and in teacher exposure to specialized professional development, particularly in maker education (Lee et al., 2020). The limited knowledge of accessibility guidelines among educational staff (Gutiérrez Y Restrepo & Floris, 2022) and inadequate Assistive Technology literacy (Shaheen, 2024) point to a pressing need for enhanced professional development.

Student engagement and interaction

The dynamics of student engagement in SLE contexts present unique challenges, particularly in maintaining consistent participation across diverse student populations. Younger children often struggle with maintaining engagement (Almourad & Bataineh, 2020), while mixed-ability classrooms face challenges in ensuring balanced participation between sighted and visually impaired students (Neto et al., 2021; Rocha et al., 2023). The complexity increases in unstructured activities, where students with disabilities may face additional barriers to participation (Lee et al., 2020). Additionally, studies highlight difficulties in ensuring equitable engagement between neurodivergent and neurotypical children (Sobel et al., 2024) and in overcoming nonverbal communication barriers (Neto et al., 2021).

Research and implementation limitations

The empirical foundation for SLE efficacy faces several methodological constraints that impact assessment validity. These limitations include insufficient sample sizes affecting generalizability (Cunha-Pérez et al., 2024; Dinét et al., 2022), truncated study durations due to external variables such as COVID-19 (Fung et al., 2023), and dependence on self-reported data potentially introducing response bias (Ayu Anggraini et al., 2021). Findings indicate that group composition heterogeneity and potential novelty effects in technology adoption contribute to challenges in ecological validity within educational settings (Dinét et al., 2022; Garzotto et al., 2020; Neto et al., 2020).

Best practices and recommendations for inclusive SLEs (RQ5)

Based on the analysis of the selected studies, several best practices emerge for designing and implementing inclusive SLEs. These practices focus on technological, pedagogical, and policy-driven solutions that promote equitable access and engagement.

Technology and system design findings

A major barrier to the SLE implementation is the lack of fully accessible and adaptive technologies tailored to a diverse range of learners' needs. Studies show that incorporating multimodal accessibility capabilities, such as text-to-speech, speech-to-text, and dyslexia-friendly typefaces, and systematic integration of UDL principles into learning environments increase SLE usability (Patiño-Toro et al., 2023).

Studies document the use of AR and VR technologies to foster immersive and interactive learning experiences. These technologies enable students with disabilities to engage with content in controlled, responsible environments that are not possible in traditional digital settings (Sun et al., 2022). Ensuring compatibility with AT like screen readers and eye-tracking systems, has been reported as an important aspect of accessible SLE design.

Additionally, findings highlight the development of intuitive and user-friendly interfaces, emphasizing the importance of simplifying navigation structures, providing adjustable text sizes, and incorporating customizable interaction modes. Co-designing these interfaces with students with disabilities ensures that accessibility challenges are addressed from a user-centered perspective, leading to more effective and inclusive learning environments (Sulek et al., 2019).

Inclusive learning findings

Studies indicate that many educators lack adequate training to implement inclusive digital learning strategies, emphasizing the need for targeted professional development (Passmore et al., 2023; Sulek et al., 2019). Training programs should include hands-on experience with assistive technologies, guidance on inclusive instructional design, and strategies for fostering student engagement in digital environments (Piki & Markou, 2023).

Studies advocate the use of structured training frameworks in which professional development is provided as ongoing learning rather than one-time seminars (Canet & Tonidandel, 2022). Furthermore, incorporating collaborative learning models has been proven to improve the deployment of accessibility elements in SLEs (Amaral et al., 2021).

Institutional and policy findings

Several studies highlight the need for clear institutional guidelines, funding for accessibility initiatives, and structured policies that promote equitable learning environments (Gutiérrez Y Restrepo & Floris, 2022; Kearney-Volpe et al., 2023). Research suggests that universities and schools should follow international guidelines such as the Web Content Accessibility Guidelines (WCAG) to ensure that SLE platforms meet established accessibility benchmarks (Pal et al., 2023). Additionally, accessibility audits and compliance assessments should be conducted regularly to identify and address usability barriers that may hinder students with disabilities from fully engaging with learning platforms.

Discussion

In this section, we summarize our SLR's findings, emphasizing both the potential and constraints of SLEs for supporting students with disabilities. It also investigates technological constraints, pedagogical issues, institutional and policy consequences, and future research initiatives to improve the inclusivity of SLEs.

Trends in Inclusive SLE Research

This SLR highlights a growing body of research on SLE, demonstrating their potential to significantly enhance inclusiveness in educational settings. Notably, the rapid increase in research activity, especially peaking in 2023, signals increased recognition of the role of technology in supporting inclusive education (Cheung et al., 2021a; El-Sabagh, 2021). Momentum in this field has been driven by the proliferation of technologies such as AR, IoT, and learning analytics. Recent trends indicate a shift toward real-world applications and interdisciplinary collaborations (Gao & Ismail, 2025), with inclusive digital learning emerging as a central theme in educational technology scholarship, particularly in high-income and university-based contexts (Varriale et al., 2022). Our review also found that most contributions are journal articles or conference proceedings in educational technology, accessibility, or interdisciplinary computing venues, accompanied by a steady increase in cross-disciplinary collaborations.

Recent bibliometric studies also confirm this trajectory. A 20-year analysis by Saini et al. (2025) shows increased global output and interest in inclusive educational technologies. Moreover, research intersections with AI and accessibility frameworks are becoming dominant themes in smart learning, as shown in recent innovations like the Smart Learning Assistance Tool (Srivastava et al., 2021).

Disability Focus and Educational Level

The identified tools and technologies address a wide range of disabilities, with a special emphasis on ASD, hearing impairment, vision impairment, and learning disabilities. SLEs can personalize learning experiences through a variety of applications, including AR, robotics, IoT-enabled classrooms, and adaptive e-learning platforms (Herrera Poma et al., 2023; Neto et al., 2023; Patiño-Toro et al., 2023). While these findings support the potential of SLEs in inclusive education, they also highlight substantial research gaps (Garzotto et al., 2020; Lobo et al., 2020). Technologies for students with physical and intellectual disabilities remain underexplored compared to sensory and cognitive impairments (AlTaleb et al., 2024; Cunha-Pérez et al., 2024). This highlights the need for targeted research and technological development addressing these underexplored disability categories. Accessibility and usability remain major concerns, revealing substantial gaps between theoretical potential and practical effectiveness (Piki & Markou, 2023). Without addressing these fundamental issues, the full benefits of SLEs may remain limited, highlighting the need for rigorous real-world validation studies (Fung et al., 2023). Ecologically, this uneven emphasis across disability categories reveals a research imbalance: sensory and cognitive disabilities are supported by design and academic affordances, whereas systemic and social affordances for physical and intellectual disabilities remain limited. Recent literature shows that faculty familiarity and systemic barriers reinforce the exclusion of physical and sensory disabilities, especially at the university level (Saadun et al., 2024). Faculty awareness and experience also play a role in shaping inclusion scope (Alqarni, 2021).

Methodologies and Technological Applications

This review indicates that most studies utilize experimental or quasi-experimental designs, usability testing, or design-based research approaches. Applications include

wearable devices, personalized e-learning platforms, real-time feedback tools, and AR/VR simulations tailored to support independent learning and accessibility. Participatory and co-design methodologies are gaining attention, particularly those involving people with disabilities in the development process (Pernia & Salmón, 2020). AI technologies are increasingly being used to enhance accessibility and personalization in SLEs. Adaptive tools such as speech recognition, NLP, and personalized content delivery are helping remove barriers for students with disabilities (Kunda et al., 2025). Additionally, descriptive analytics dashboards are empowering educators to make data-informed adjustments to instructional design and learner support, especially for marginalized learners (Costas-Jauregui et al., 2021). These methodological patterns reinforce the dominance of design and academic affordances in current SLE research. The ecological model, however, highlights the need to also examine how such technologies interact with systemic affordances (e.g., policy frameworks) and pedagogical affordances (e.g., teacher knowledge). Without this integration, innovation risks remaining tool-centered rather than learner-centered.

Challenges in adoption and Integration

This systematic review identifies significant technological challenges that hinder the widespread adoption of SLEs. Among these are hardware limitations and resource limits, which prevent institutions from fully incorporating assistive technologies into their learning environments (Kaimara et al., 2021; Kearney-Volpe et al., 2021). Moreover, technical hurdles like inconsistent performance in varying environmental conditions (e.g., lighting, audio quality) negatively affect usability (Canet & Tonidandel, 2022; Sun et al., 2022). These findings align with previous research that emphasizes the necessity for robust, adaptable infrastructure to ensure effective implementation (Zhang et al., 2024).

Design limitations in inclusive SLEs are increasingly linked to poor integration of Universal Design for Learning (UDL) principles. UDL frameworks provide proactive strategies that can accommodate diverse learning needs, yet their implementation faces barriers such as inadequate training and infrastructure (Wahyuni et al., 2025). Additionally, the application of UDL in online and blended instructional design has been shown to increase flexibility and engagement for all learners (Rao, 2021).

While technical developments are important, their impact depends on successful pedagogical integration. Educators' proficiency and preparedness significantly influence their successful adoption (Passmore et al., 2023; Shaheen, 2024). The evidence indicates substantial gaps in teachers' training concerning accessibility standards, adaptive instruction, and effective use of assistive technologies (Piki & Markou, 2023; Van Der Steen et al., 2020). These challenges resonate with the broader literature that advocates comprehensive, targeted training and professional development as essential for effective pedagogical integration (Almourad & Bataineh, 2020; Amaral et al., 2021).

A lack of training resources and institutional incentives further prevents inclusive tech from scaling (Kamran et al., 2023). Systemic policies remain fragmented across institutions and lack enforcement (Karisa et al., 2024). Even programs like Brazil's "Include" initiative show that without institutional commitment and central accessibility centers, policy efforts remain isolated and unsustainable (Anicésio & Anache, 2024).

These challenges demonstrate that while inclusive SLEs are advancing in terms of design affordances, systemic affordances remain fragmented and weak. The absence of coherent institutional policies, insufficient teacher preparation, and uneven accessibility standards reflect gaps at the system and practice levels of the ecology. Addressing these dimensions is essential to move beyond isolated innovations toward sustainable inclusion.

Best Practices for implementing inclusive SLEs

To address these challenges, future implementations should adhere strictly to UDL principles, enhancing adaptability and ensuring compatibility with assistive technologies (Elmahmoudi et al., 2021; Sun et al., 2022). Specifically, institutions should prioritize cross-platform accessibility to accommodate diverse learners' needs and cloud-based solutions to reduce hardware dependency and improve scalability in resource-constrained settings.

Current research highlights several promising practices. Aftab et al. (2024) advocate for co-teaching, assistive personalization, and learner-centered curriculum design as foundational to equitable SLE adoption. The role of educator training is also critical. Institutions should prioritize equipping educators with the skills and knowledge to implement adaptive learning technologies, UDL principles, and accessible digital tools through professional development programs that provide hands-on training, continuous feedback, and support systems to sustain inclusive practices (Larios & Zetlin, 2023).

Social-emotional support also contributes to inclusive success. Recent evidence highlights peer mentorship as a valuable strategy for supporting students with disabilities in inclusive settings. For instance, Maia et al. (2024) found that structured Peer Mentoring Circles fostered academic engagement, reduced isolation, and enhanced inclusion for students with intellectual disabilities in higher education.

Best practices for inclusive SLE implementation require a multifaceted approach, combining adaptive technologies, pedagogical training, and user-centered design. Prioritizing Universal Design for Learning (UDL) principles, investing in educator capacity-building, and involving learners with disabilities in co-design processes have emerged as effective strategies (Rao, 2021; Wahyuni et al., 2025).

In addition to technological and pedagogical considerations, sustainable implementation of inclusive SLEs depends heavily on institutional and policy support. Despite advancements in assistive and adaptive technologies, the absence of standardized accessibility policies and uneven adoption of international guidelines (e.g., WCAG), restricts comprehensive integration (Gutiérrez Y Restrepo & Floris, 2022; Kukushkina et al., 2022). The results advocate strongly for institutional policies enforcing universal accessibility standards, ensuring all digital platforms within educational institutions are consistently inclusive and equitable (Fung et al., 2023; Kearney-Volpe et al., 2023). Without systemic policy-driven frameworks, inclusive practices may remain isolated and unsustainable, limiting the long-term impact of SLE initiatives. Taken ecologically, best practices show that sustainable SLEs require layered affordances: design (adaptive tech), pedagogical (faculty training), social (peer mentoring), and systemic (policy enforcement).

Implications for future research and practice

The methodological limitations identified in this review highlight the need for future research to prioritize larger-scale, longitudinal, and ecologically valid studies (Cunha-Pérez et al., 2024; Dinet et al., 2022). Future research should perform extensive empirical testing of SLEs in real-world educational contexts over extended periods, which will allow a deeper understanding of long-term impacts and sustained usage among students with disabilities. Moreover, incorporating mixed-methods approaches can yield richer, more actionable insights by combining quantitative rigor with qualitative depth. This enables researchers to evaluate both measurable outcomes and user experiences, capture diverse needs within the disability community, and support iterative design based on authentic, context-sensitive data (Larios & Zetlin, 2023; Rocha et al., 2023).

Theoretical implications include expanding the application of UDL by integrating emerging technologies such as AI, IoT, and emotion-aware systems to support new models of inclusive pedagogy. This review encourages scholars to develop frameworks that connect personalization algorithms with accessibility goals, an under-theorized intersection in inclusive education literature.

Practical implications are evident in how SLEs can enable educators to adopt multimodal teaching approaches that meet varied learner needs. For example, descriptive analytics tools now allow instructors to dynamically adjust content and pace, enhancing real-time responsiveness to students with disabilities (Costas-Jauregui et al., 2021). Institutions can also use these findings to inform procurement decisions, ensuring that selected learning tools align with accessibility standards and interoperability goals.

Managerial and policy-level implications emphasize the importance of institutional frameworks that embed accessibility into digital strategy. Educational leaders should establish centralized accessibility oversight bodies and enforce standards like WCAG across all digital learning platforms. As shown by Kameni and Koumédio (2023), inclusive education policy in smart learning ecosystems is most effective when supported by interdepartmental coordination and clear accountability measures (Kameni & Koumédio, 2023).

Additionally, co-design approaches, where students with disabilities actively contribute to system development, should be adopted at the policy level as a standard for inclusive innovation (Yu, 2024). Managerial stakeholders must also invest in ongoing faculty training, change management, and resource allocation strategies that support long-term accessibility implementation. The ecological framework reinforces these implications by highlighting that future research should systematically examine interactions across affordances rather than isolated interventions. For instance, emotion-aware AI (design affordance) must be studied alongside teacher expertise (pedagogical affordance) and policy enforcement (systemic affordance) to evaluate whether inclusive outcomes are sustained. Ecologically valid research of this kind will allow SLEs to transition from promising prototypes to embedded inclusive systems.

Conclusion and Future Recommendations

This SLR paper has provided a comprehensive analysis of the role of smart learning environments in fostering inclusive education for students with disabilities. By synthesizing findings from 43 selected peer-reviewed studies published between 2014 and 2024, it provides a comprehensive analysis of SLEs. It highlights the key technological advancements, the existing challenges, and the best practices for designing and implementing accessible and inclusive SLEs. The findings demonstrate the potential of SLEs to enhance education through emerging technologies and interactive digital tools. These innovations enable more personalized and adaptive learning experiences tailored to the diverse students' needs. For instance, Smart Learning Assistance (SLA) tools that integrate speech-to-text conversion, sign language recognition, and Braille-to-text communication have been shown to significantly enhance accessibility and interaction in the classroom. Additionally, research suggests that inclusive education environments promote social integration and cognitive development for students with disabilities, particularly when supported by innovative pedagogical frameworks.

However, despite these promising advancements, significant challenges remain. These include technical barriers such as inadequate infrastructure, limited assistive technology integration, and inconsistent accessibility standards. Additionally, systemic issues, including insufficient educator training, a lack of standardized accessibility policies, and limited research on long-term efficacy, further restrict equitable learning experiences for learners with disabilities.

Ensuring inclusivity in digital education requires a coordinated approach that integrates technological, pedagogical, policy-driven, and infrastructure-focused strategies. Based on this review, the following areas are particularly critical.

- SLE designers should prioritize adaptive and multimodal accessibility solutions aligned with Universal Design for Learning (UDL) principles. Recent work highlights the effectiveness of AI-powered tools that dynamically adjust content delivery based on learners' cognitive and emotional profiles (Ahmed et al., 2025).
- Institutions should enhance professional development and continuous training programs focused on inclusive pedagogical practices. Faculty preparedness, empathy, and practical understanding of accessibility tools have a direct impact on implementation success (Moriña & Orozco, 2020; Sánchez-Díaz et al., 2025).
- Educational institutions must adopt and enforce comprehensive digital accessibility standards, supported by clearly articulated policies and dedicated resources. The absence of systemic policy enforcement and WCAG adoption continues to fragment inclusive practices in higher education (Page et al., 2023).
- Educational institutions need investment in updated technological infrastructure to overcome technical limitations impacting SLE efficacy. Cloud-based platforms and cross-platform compatibility are particularly effective for improving access in resource-constrained settings (Srivastava et al., 2021).

To address these challenges, this systematic review underscores the need for a holistic approach to SLE implementation. In this way, the integration of UDL principles remains a possible foundation to guide system development, ensuring that learning

environments cater to diverse cognitive, sensory, and motor needs. However, applying UDL in practice requires not only technical tools but cultural and institutional change, including participatory design processes that directly involve students with disabilities (Zhou et al., 2023). As a recommendation, institutions should invest in professional development programs that equip educators with the skills to integrate accessibility technologies effectively. Furthermore, policy enforcement, institutional support, and collaborative research efforts are crucial in bridging existing accessibility gaps. An emerging area deserving attention is the intersection of ethics, AI, and inclusion, especially as automated systems may unintentionally reinforce bias if designed without representative data (Morr et al., 2024).

A promising future direction for expanding the depth and applicability of this research involves incorporating expert views to complement the literature-based findings. This could include structured expert elicitation methods or focus groups to validate the relevance of identified technologies, challenges, and best practices for inclusive smart learning. Such additions would help bridge the gap between theoretical knowledge and practical application in diverse educational contexts (Ngadengon et al., 2024). Deliberative expert engagement may also reveal blind spots in current inclusion frameworks, helping anticipate unintended consequences or user misalignment.

To ensure meaningful progress, future research should focus on longitudinal studies to assess the sustained impact of SLEs on students with disabilities. More attention is also needed on underrepresented disability categories, such as intellectual and physical disabilities, to ensure truly inclusive learning environments. A further area for future research concerns the investigation of the effectiveness of adaptive technologies and personalized learning strategies to enhance accessibility and engagement for all students. Emerging work on emotion-aware AI tutors, gamified accessibility strategies, and cross-modal content delivery offers promising avenues, but these must be critically tested across real-world classroom environments (Ahmed et al., 2025). In summary, realizing the full potential of inclusive SLEs will require more than technological innovation, it will demand systemic alignment across pedagogy, policy, infrastructure, and institutional culture.

Abbreviations

AI	Artificial Intelligence
AR	Augmented Reality
ASD	Autism spectrum disorder
CRPD	United Nations Convention on the Rights of Persons with Disabilities
iMSE	Interactive Multisensory Environment
I/DD	Intellectual and developmental disabilities
IoT	Internet of Things
MOOC	Massive Open Online Courses
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
SDG4	Sustainable Development Goal 4
SLA	Smart Learning Assistance
SLE	Smart Learning Environment
SLR	Systematic literature review
UDL	Universal Design for Learning
WCAG	Web Content Accessibility Guidelines

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Author Contributions

The conception of the study and the study design was carried out by A.O., A.D., M.H.F.H., and D.A.T. Data acquisition was performed by A.D. and M.H.F.H., and the review process was conducted by A.D., M.H.F.H., and D.A.T. Interpretation of data and drafting of the initial manuscript were undertaken by A.D., M.H.F.H., and K.C. The manuscript was substantively revised by K.C. Overall supervision and critical guidance were provided by A.O. and D.A.T.

Availability of data and materials

The data that support the findings of this study are available based on a reasonable request from the corresponding author.

Declarations

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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